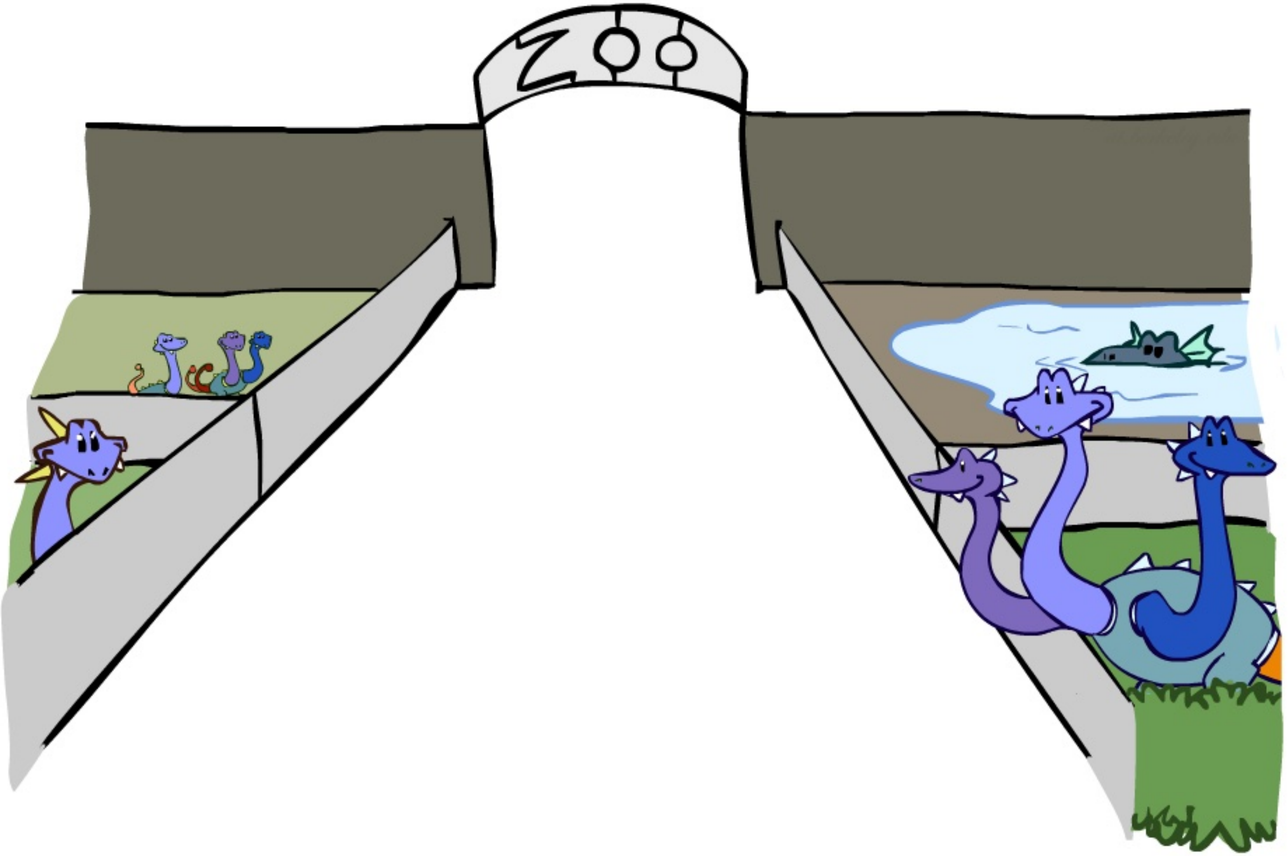
Probabilistic Reasoning

Factor Zoo



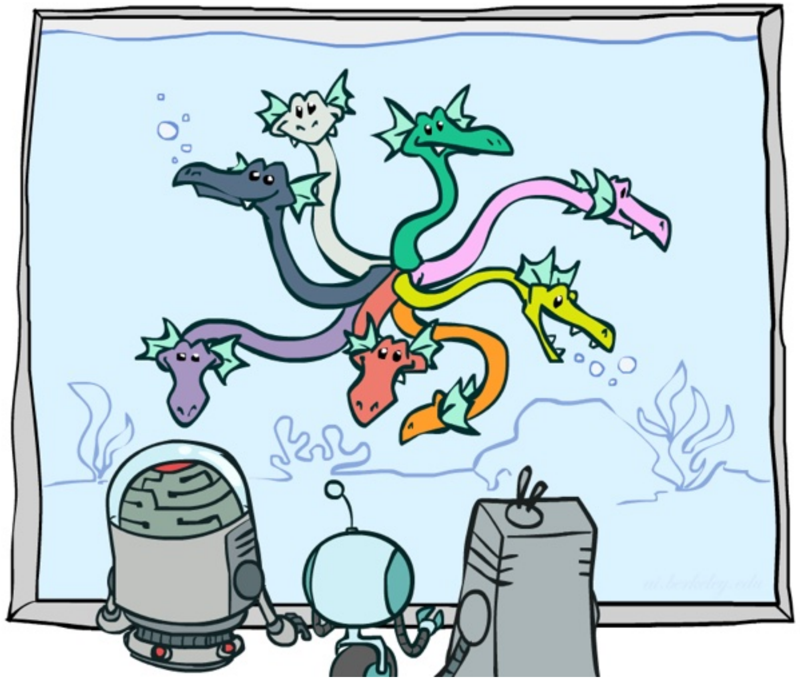
Factor Zoo I

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

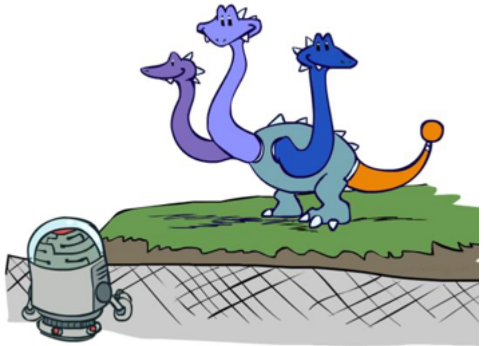
$P(cold, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3



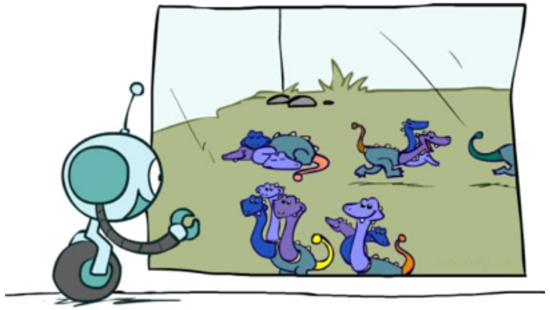
- Joint distribution: $P(X, Y)$
  - Entries $P(x, y)$ for all x, y
  - Sums to 1
- Selected joint: $P(x, Y)$
  - A slice of the joint distribution
  - Entries $P(x, y)$ for fixed x, all y
  - Sums to $P(x)$
- Number of capitals determines dimensionality of the table

Factor Zoo II



$P(W|cold)$

T	W	P
cold	sun	0.4
cold	rain	0.6

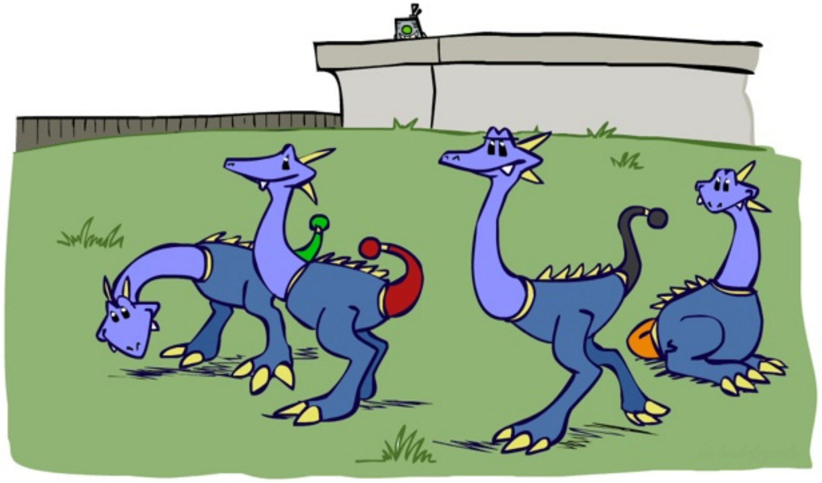


$P(W|T)$

T	W	P
hot	sun	0.8
hot	rain	0.2
cold	sun	0.4
cold	rain	0.6

- Single conditional: $P(Y \mid x)$
  - Entries $P(y \mid x)$ for fixed x, all y
  - Sums to 1
- Family of conditionals: $P(Y \mid X)$
  - Multiple conditionals
  - Entries $P(y \mid x)$ for all x, y
  - Sums to $|X|$

Factor Zoo III



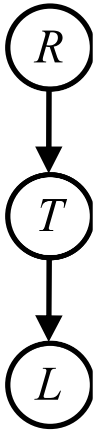
$P(rain|T)$

T	W	P
hot	rain	0.2
cold	rain	0.6

- Specified family: $P(y \mid X)$
  - Entries $P(y \mid x)$ for fixed y, but for all x
  - Sums to … who knows!

Example: Traffic Domain

- **Random Variables**
  - R: Raining
  - T: Traffic
  - L: Late for flight!
- $P(L) =$
  - $= \sum_{r,t} P(r,t,L)$
  - $= \sum_{r,t} P(r)P(t|r)P(L|t)$



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Inference by Enumeration – Procedural Outline

- Procedure tracks objects called factors
- Initial factors are local CPTs (one per node)

$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- Any known values are selected
  - For example if we know $L = true$,

$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

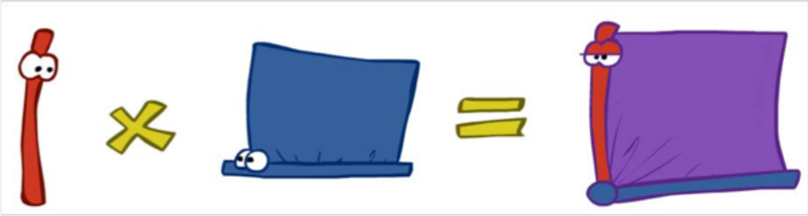
$P(+\ell|T)$

+t	+l	0.3
-t	+l	0.1

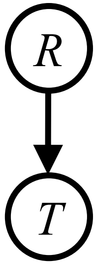
- Procedure – Join all factors and eliminate hidden variables

Operation 1 – Join Factors

- Similar to database join
- Get all factors over the joining variable
  - Build a new factor over the union of the variables involved



- Example – Join on R



$P(R)$

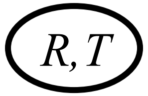
+r	0.1
-r	0.9

$\times$ $P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

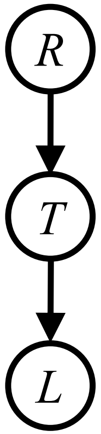
$\rightarrow$ $P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81



  - Computation of each entry is a pointwise product
    - $\forall r, t \; P(r,t) = P(r) \,.\, P(t|r)$

Example: Multiple Joins



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

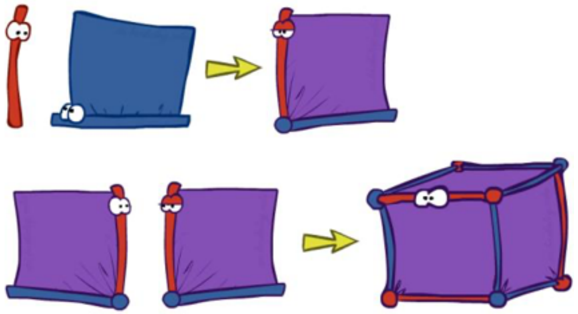
Join R

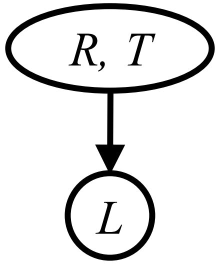
$P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

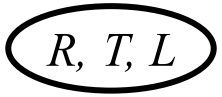
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9





Join T



$P(R,T,L)$

+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Operation 2: Eliminate

- Second basic operation: marginalization
- Take a factor and sum out a variable
  - Shrinks a factor to a smaller one
  - A projection operation
- Example:

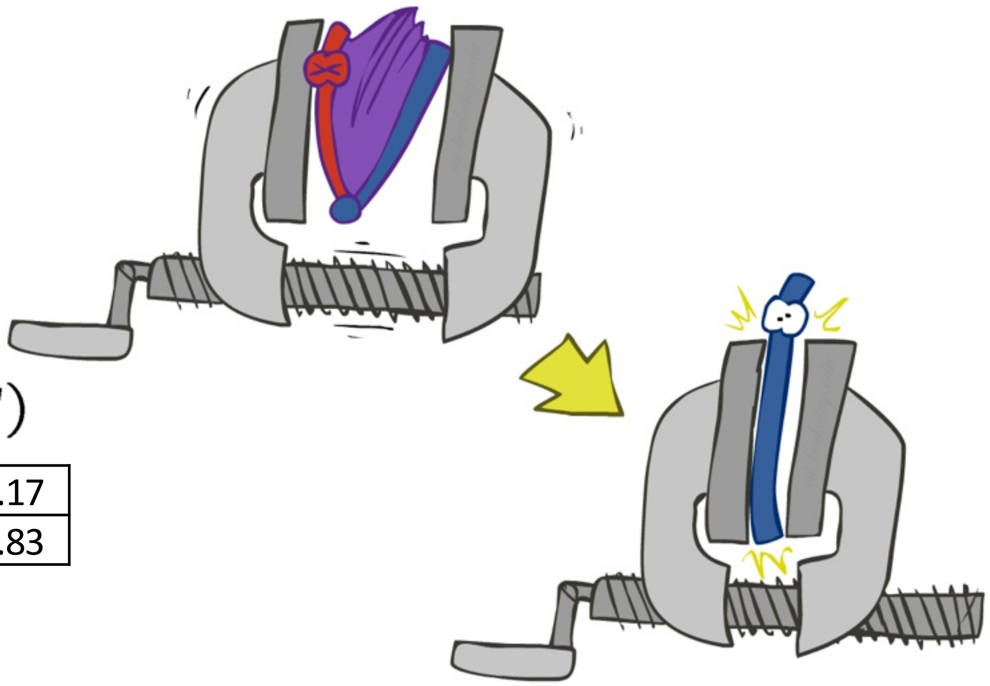
$P(R, T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

sum R →

$P(T)$

+t	0.17
-t	0.83



Multiple Elimination

R, T, L

$P(R,T,L)$

+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Sum out R $\rightarrow$

T, L

$P(T,L)$

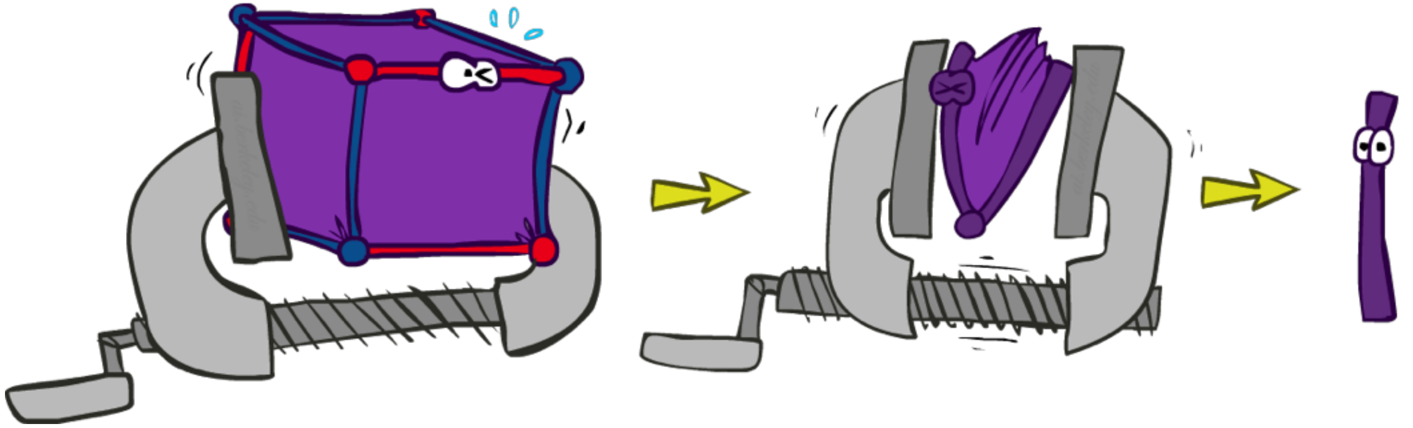
+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum out T $\rightarrow$

L

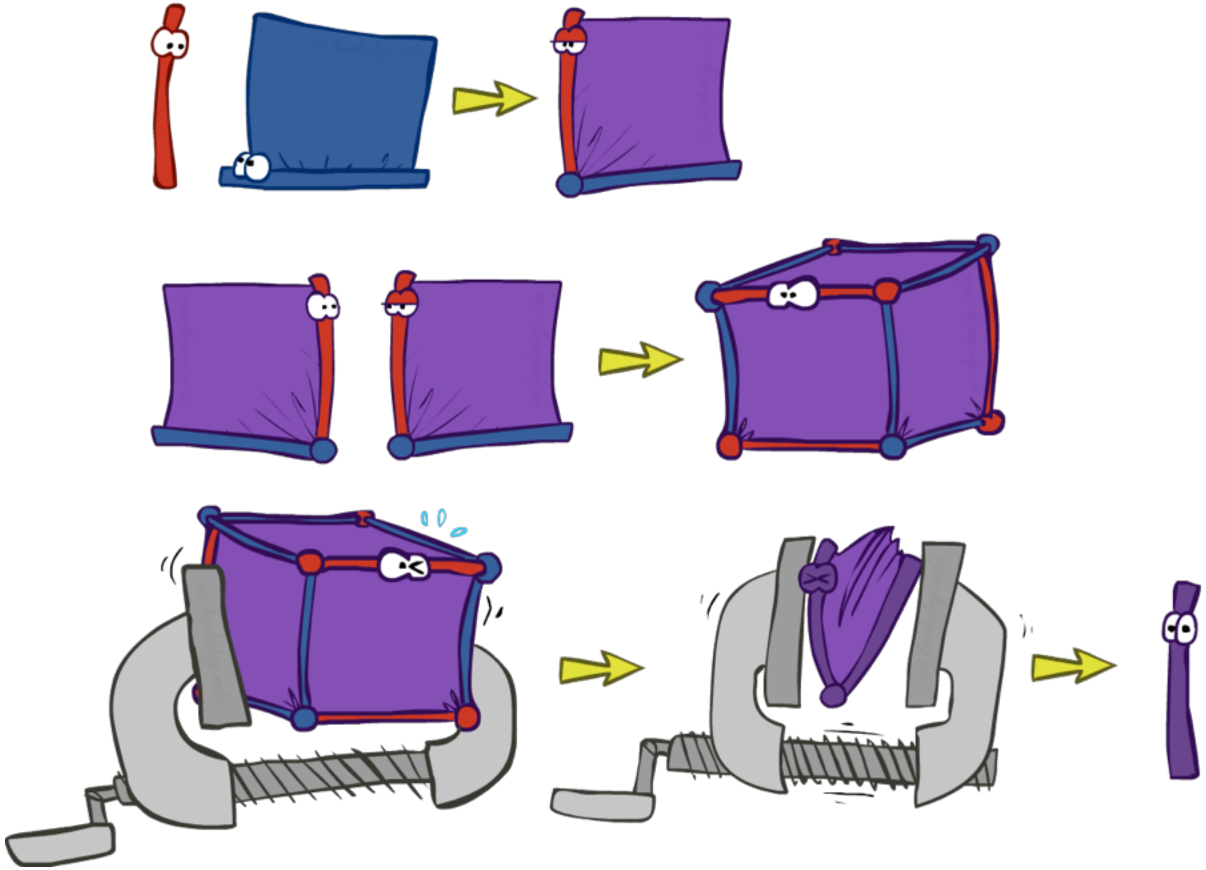
$P(L)$

+l	0.134
-l	0.866

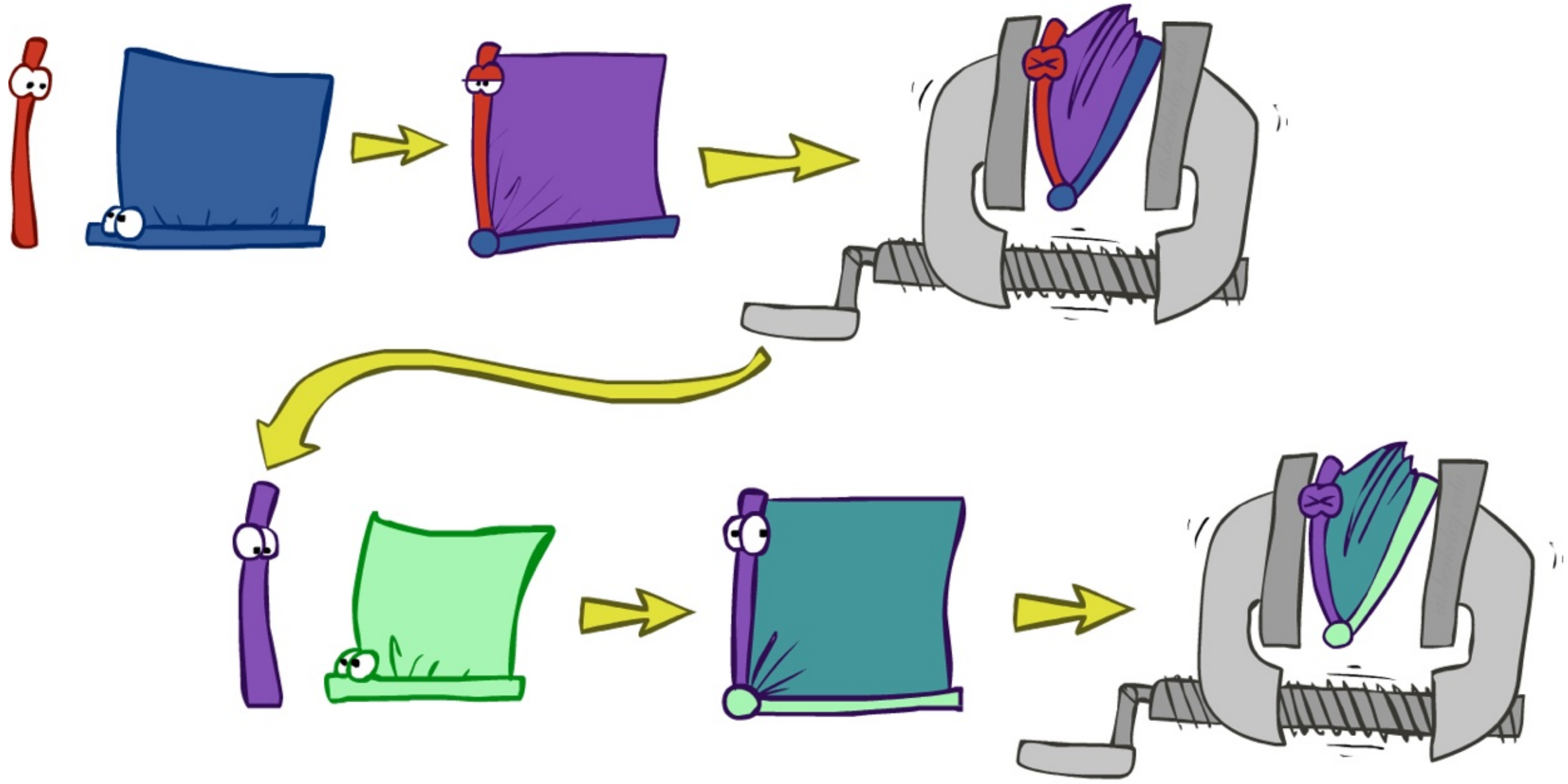


Inference by Enumeration

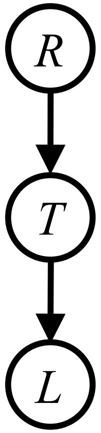
- Multiple joins and multiple eliminations



Marginalizing Early (= Variable Elimination)



Marginalizing Early! (aka VE)



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

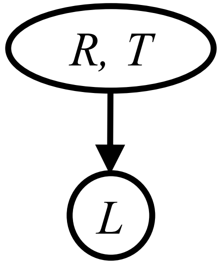
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Join R

$P(R, T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81



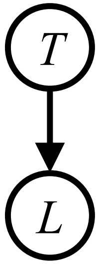
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Sum out R

$P(T)$

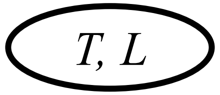
+t	0.17
-t	0.83



$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

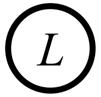
Join T



$P(T, L)$

+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum out T



$P(L)$

+l	0.134
-l	0.866

Traffic Domain

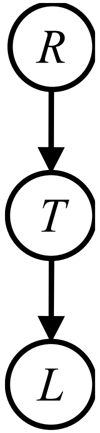
- Inference by enumeration
- $P(L) =$
  - $= \sum_{r,t} P(r,t,L)$
  - $= \sum_{r,t} P(L|t)P(r)P(t|r)$

Join on r

Join on t

Eliminate r

Eliminate t



- Variable elimination
- $P(L) =$
  - $= \sum_{r,t} P(r,t,L)$
  - $= \sum_{t} P(L|t) \sum_{r} P(r)P(t|r)$

Join on r

Eliminate r

Join on t

Eliminate t

Evidence I

- If evidence, start with factors that select that evidence
  - No evidence uses these initial factors:

$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- Computing $P(L|r)$, the initial factors become:

$P(+r)$

+r	0.1

$P(T|+r)$

+r	+t	0.8
+r	-t	0.2

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- We eliminate all variables other than query + evidence

Evidence II

- Result will be a selected joint of query and evidence
  - E.g. for $P(L \mid r)$, we would end up with:

$P(+r, L)$

+r	+l	0.026
+r	-l	0.074

Normalize

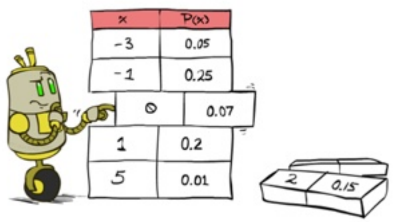
$P(L| + r)$

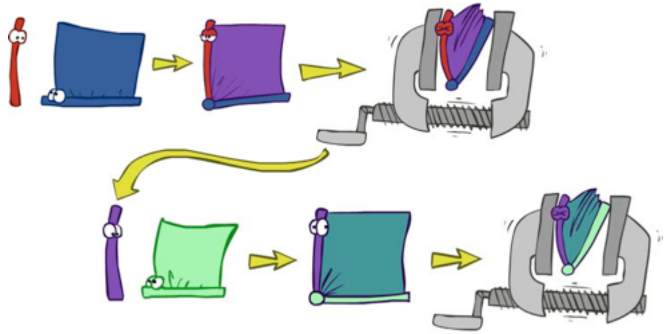
+l	0.26
-l	0.74

- To get our answer, just normalize this!
- That 's it!

General Variable Elimination

- Query: $P(Q|e_1, \ldots, e_k)$
- Start with initial factors:
  - Local CPTs (but instantiated by evidence)
- While there are still hidden variables (not Q or evidence):
  - Pick a hidden variable H
  - Join all factors mentioning H
  - Eliminate (sum out) H
- Join all remaining factors and normalize

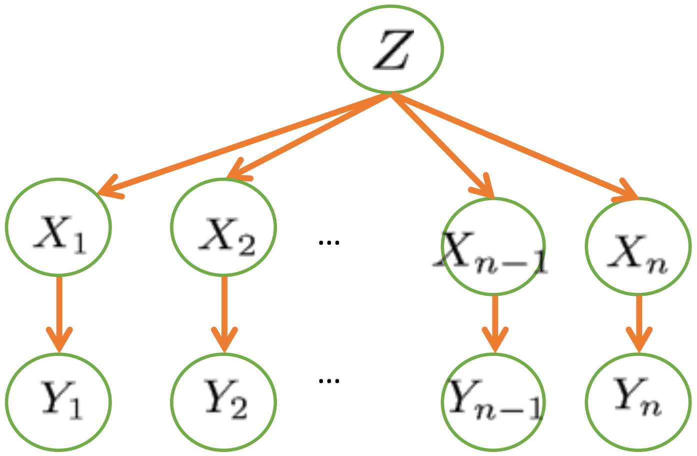




$\times \frac{1}{Z}$

Variable Elimination Ordering

- For the query $P(X_n|y_1, \ldots, y_n)$ work through the following two different orderings as done in previous slide: $Z, X_1, \ldots, X_{n-1}$ and $X_1, \ldots, X_{n-1}, Z$. What is the size of the maximum factor generated for each of the orderings?



Another Variable Elimination Example

Query: $P(X_3|Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$

Start by inserting evidence, which gives the following initial factors:

$$P(Z), P(X_1|Z), P(X_2|Z), P(X_3|Z), P(y_1|X_1), P(y_2|X_2), P(y_3|X_3)$$

Eliminate X_1, this introduces the factor $f_1(y_1|Z) = \sum_{x_1} P(x_1|Z)P(y_1|x_1)$, and we are left with:

$$P(Z), P(X_2|Z), P(X_3|Z), P(y_2|X_2), P(y_3|X_3), f_1(y_1|Z)$$

Eliminate X_2, this introduces the factor $f_2(y_2|Z) = \sum_{x_2} P(x_2|Z)P(y_2|x_2)$, and we are left with:

$$P(Z), P(X_3|Z), P(y_3|X_3), f_1(y_1|Z), f_2(y_2|Z)$$

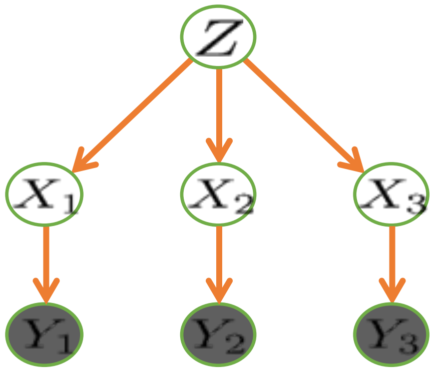
Eliminate Z, this introduces the factor $f_3(y_1, y_2, X_3) = \sum_z P(z)P(X_3|z)f_1(y_1|Z)f_2(y_2|Z)$, and we are left with:

$$P(y_3|X_3), f_3(y_1, y_2, X_3)$$

No hidden variables left. Join the remaining factors to get:

$$f_4(y_1, y_2, y_3, X_3) = P(y_3|X_3), f_3(y_1, y_2, X_3)$$

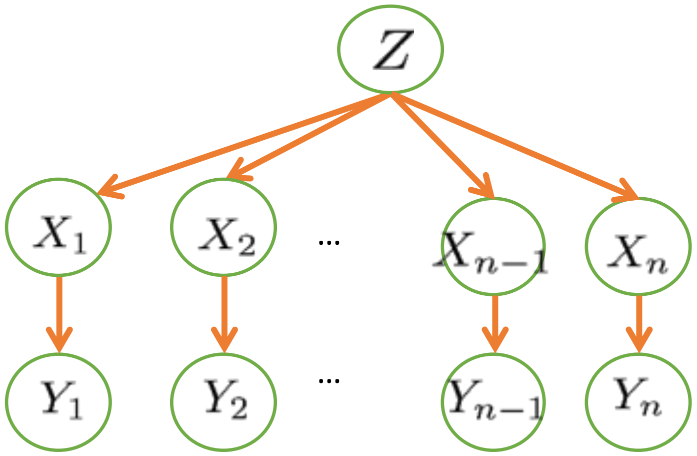
Normalizing over X_3 gives $P(X_3|y_1, y_2, y_3) = f_4(y_1, y_2, y_3, X_3) / \sum_{x_3} f_4(y_1, y_2, y_3, x_3)$



Computational complexity critically depends on the largest factor being generated in this process. Size of factor = number of entries in table. In example above (assuming binary) all factors generated are of size 2 --- as they all only have one variable (Z, Z, and X_3 respectively).

Variable Elimination Ordering

- For the query $P(X_n|y_1, \ldots, y_n)$ work through the following two different orderings as done in previous slide: $Z, X_1, \ldots, X_{n-1}$ and $X_1, \ldots, X_{n-1}, Z$. What is the size of the maximum factor generated for each of the orderings?
  - Answer: 2^n versus 2 (assuming binary)
- In general: the ordering can greatly affect efficiency.



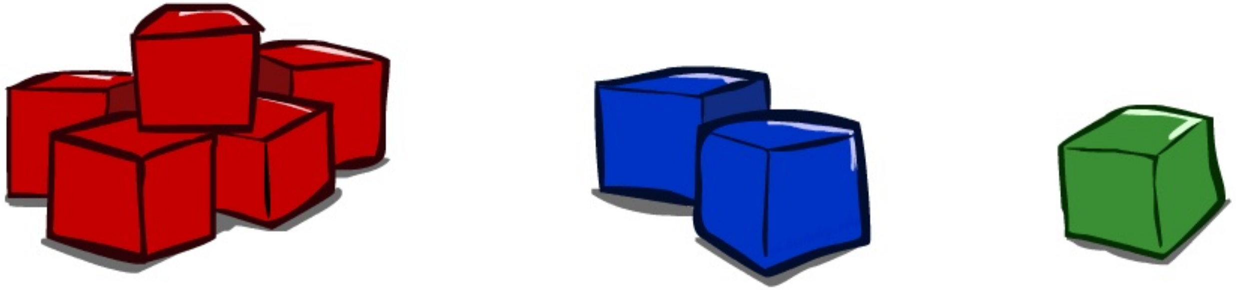
VE: Computational and Space Complexity

- The computational and space complexity of variable elimination is determined by the largest factor
- The elimination ordering can greatly affect the size of the largest factor.
- Does there always exist an ordering that only results in small factors?
  - No!
- Exact inference in BNs is NP-Hard

Outline

- Probability theory - basics
- Probabilistic Inference
  - Inference by enumeration
- Conditional Independence
- Bayesian Networks
- Inference in Bayesian Networks
  - Exact Inference
  - Approximate Inference

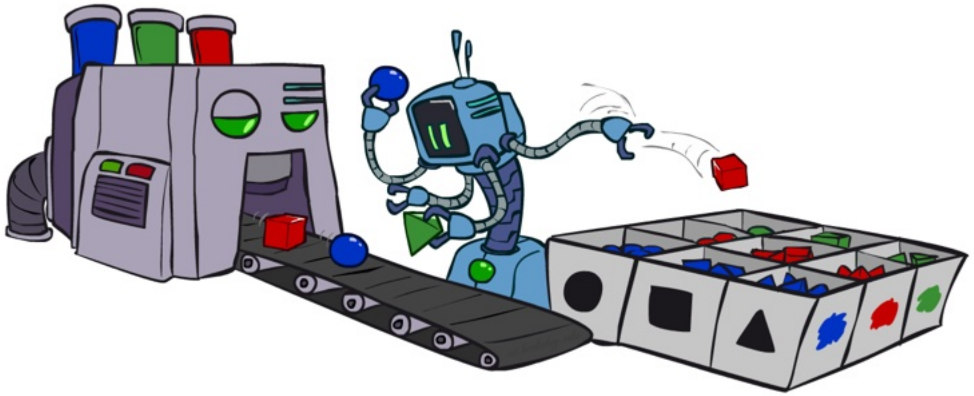
Approximate Inference: Sampling



Sampling

- **Basic idea**
  - Draw N samples from a sampling distribution S
  - Compute an approximate posterior probability
  - Show this converges to the true probability P

- **Why sample?**
  - Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



Sampling Basics

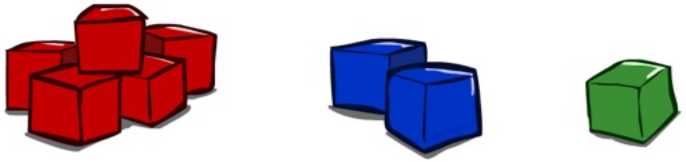
- Sampling from given distribution
  - Step 1: Get sample u from uniform distribution over [0, 1)
  - E.g. random() in python

C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \le u < 0.6, \rightarrow C = red$$
$$0.6 \le u < 0.7, \rightarrow C = green$$
$$0.7 \le u < 1, \rightarrow C = blue$$

- Step 2: Convert this sample u into an outcome for the given distribution by having each outcome associated with a sub-outcome associated with a sub-interval of [0,1) with sub-interval size equal to probability of the outcome

- If random() returns *u* = 0.83, then our sample is *C* = blue
- E.g, after sampling 8 times:



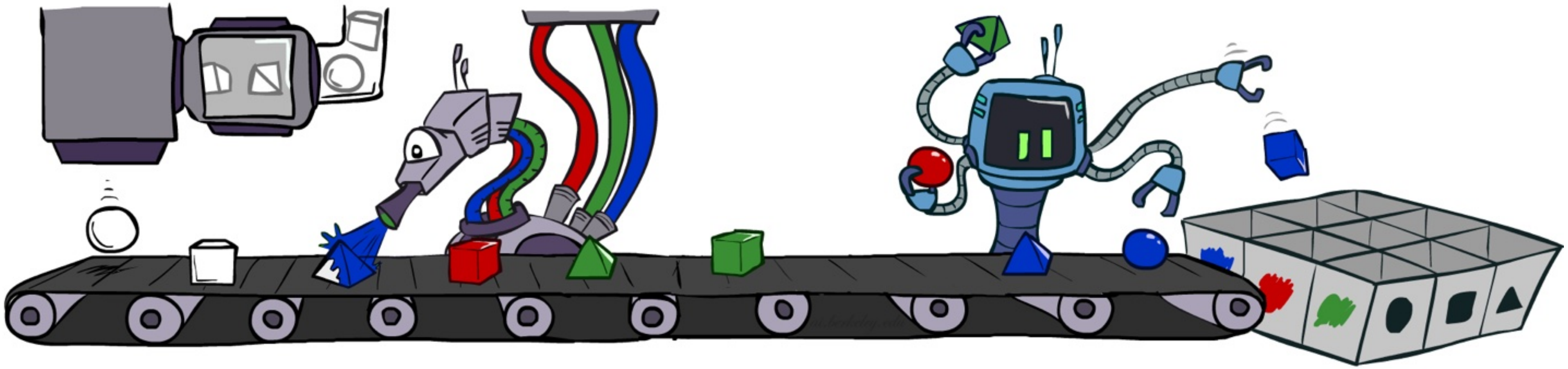
Sampling in Bayes' Nets

- Prior Sampling (Direct Sampling)
- Rejection Sampling

Prior (Direct) Sampling

- Ignore evidence. Sample from the joint probability.
- Do inference by counting the right samples.

- For $i = 1, 2, \ldots, n$
  - Sample x_i from $P(Xi \mid Parents(Xi))$
- Return $(x_1, x_2, \ldots, x_n)$



Prior Sampling

$P(C)$

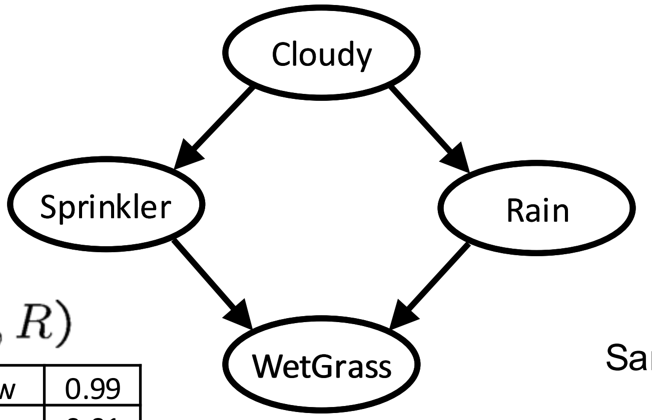
+c	0.5
-c	0.5

$P(S|C)$

+c	+s	0.1
	-s	0.9
-c	+s	0.5
	-s	0.5

$P(R|C)$

+c	+r	0.8
	-r	0.2
-c	+r	0.2
	-r	0.8



$P(W|S,R)$

+s	+r	+w	0.99
		-w	0.01
	-r	+w	0.90
		-w	0.10
-s	+r	+w	0.90
		-w	0.10
	-r	+w	0.01
		-w	0.99

Samples:

+c, -s, +r, +w

-c, +s, -r, +w

…

Example

- We'll get a bunch of samples from the BN:

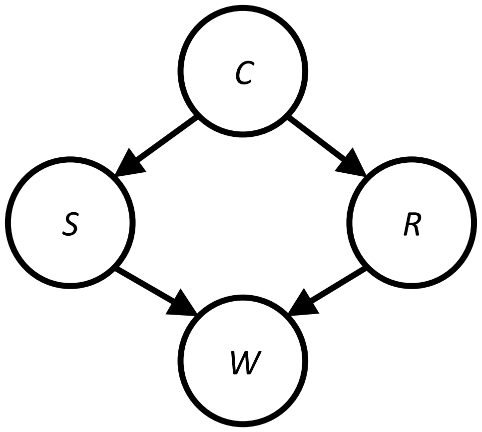
  +c, -s, +r, +w

  +c, +s, +r, +w

  -c, +s, +r, -w

  +c, -s, +r, +w

  -c, -s, -r, +w



- If we want to know $P(W)$
  - We have counts <+w:4, -w:1>
  - Normalize to get P(W) = <+w:0.8, -w:0.2>
  - This will get closer to the true distribution with more samples
  - Can estimate anything else, too
  - What about $P(C|+w)$? $P(C|+r,+w)$? $P(C|-r,-w)$?

Prior Sampling Analysis

- This process generates samples with probability:

$$S_{PS}(x_1, \dots, x_n) = \prod_{i=1}^{n} P\big(x_i | Parents(x_i)\big) = P(x_1, \dots, x_n)$$

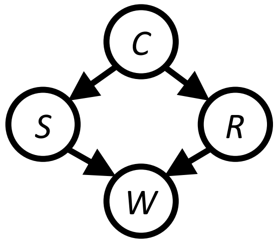
- …i.e. the BN's joint probability
- Let the number of samples of an event be $N_{PS}(x_1, \dots, x_n)$
- Then $\lim_{N\to\infty} \hat{P}(x_1, \dots, x_n) = \lim_{N\to\infty} N_{PS}(x_1, \dots, x_n)/N$

$$= S_{PS}(x_1, \dots, x_n) = \mathrm{P}(x_1, \dots, x_n)$$

- I.e., the sampling procedure is consistent

Rejection Sampling

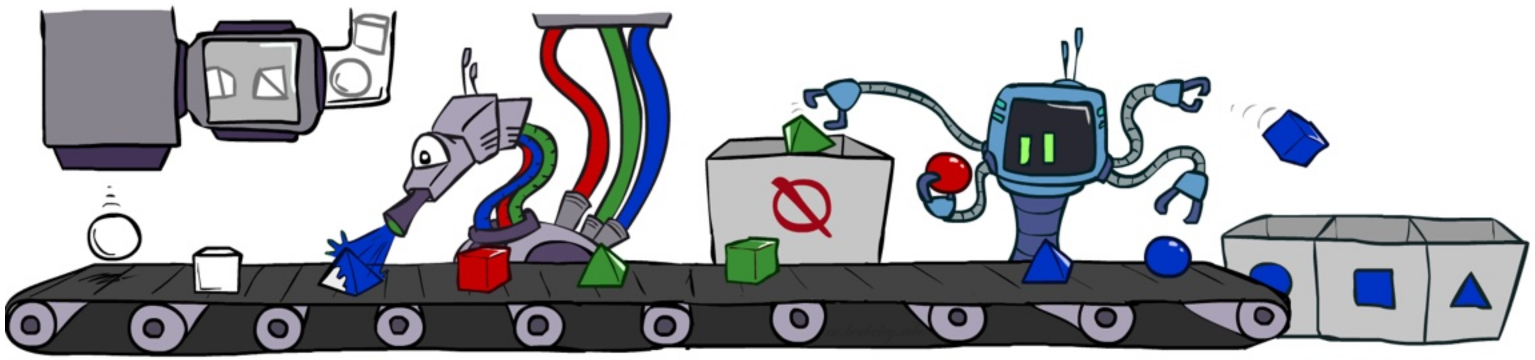
- Let's say we want $P(C|+s)$
  - Tally C outcomes, but ignore (reject) samples which don't have $S = +s$
  - This is called rejection sampling
  - It is also consistent for conditional probabilities (i.e., correct in the limit)



+c, -s, +r, +w
+c, +s, +r, +w
-c, +s, +r, -w
+c, -s, +r, +w
-c, -s, -r, +w

Rejection Sampling

- Evidence instantiation
- For $i = 1, 2, \ldots, n$
  - Sample x_i from $P(Xi \mid Parents(Xi))$
  - If x_i not consistent with evidence
  - Reject: Return, and no sample is generated in this cycle
- Return $(x_1, \ \ldots, x_n)$



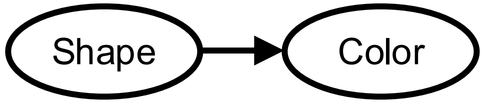
Rejection Sampling Analysis

- Let $\hat{P}(X|e)$ be the estimated distribution that the algorithm returns.
- By definition: $\hat{P}(X|e) = \alpha N_{PS}(X, e) = \frac{N_{PS}(X,e)}{N_{PS}(e)}$
- We already know that
  - $N_{PS}(x_1, \dots, x_n)/N \approx \mathrm{P}(x_1, \dots, x_n)$ and
  - $N_{PS}(x_1, \dots, x_m)/N \approx \mathrm{P}(x_1, \dots, x_m)$ for $m < n$
- Thus $\hat{P}(X|e) \approx P(X|e)/P(e) = P(X|e)$
- Sampling procedure is consistent

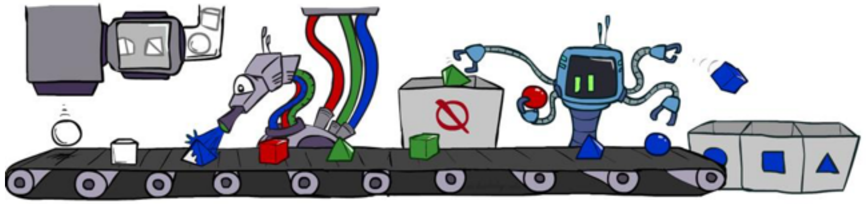
Rejection Sampling

Drawback?

- If evidence is unlikely, rejects lots of samples
- Evidence not exploited as you sample
- Consider $P(Shape|blue)$



~~pyramid, green~~
~~pyramid, red~~
sphere, blue
~~cube, red~~
~~sphere, green~~



Outline

- ❑ Probability theory - basics
- ❑ Probabilistic Inference
  - o Inference by enumeration
- ❑ Conditional Independence
- ❑ Bayesian Networks
- ❑ Inference in Bayesian Networks
  - o Exact Inference
  - o Approximate Inference